

# Demo File

*Keywords:*

## 1 Install Docker

```
# Ubuntu/Debian
sudo apt-get update
sudo apt-get install docker.io
sudo systemctl start docker
sudo systemctl enable docker

# Add your user to docker group (to run without sudo)
sudo usermod -aG docker $USER
newgrp docker

# Verify Docker is working
docker --version
docker ps
```

## 2 Pull GitHub MCP Server Docker Image

Pull docker image using,

```
docker pull ghcr.io/github/github-mcp-server
```

Verify the downloaded images by,

```
docker images | grep github-mcp-server
```

## 3 Get GitHub Personal Access Token

1. Go to: <https://github.com/settings/tokens>
2. Click "Generate new token (classic)"
3. Give it a name: "MCP Server Token"
4. Select scopes:
  - (a) repo (Full control of private repositories)
  - (b) read:user (Read user profile data)
  - (c) read:org (Read org data - optional)

5. Click "Generate token" and copy

## 4 Set Environment Variable

```
# Set GITHUB_TOKEN
export GITHUB_TOKEN='ghp_your_token_here'

# Verify
echo $GITHUB_PERSONAL_ACCESS_TOKEN
```

## 5 Test the MCP Server Manually

```
# Test the Docker container directly
docker run -i --rm \
-e GITHUB_PERSONAL_ACCESS_TOKEN \
ghcr.io/github/github-mcp-server
```

## 6 Code Output

Run: python main.py. You should see,

```
=====
MCP Application - REAL GitHub + FileSystem
=====

=====
Available Personas:
=====
eng01: Alex Chen (Engineering) - Software Engineer
      GitHub: read_write
      FileSystem: Engineering

it01: Priya Nair (IT) - ITSM Analyst
      GitHub: read_only
      FileSystem: IT

sales01: Marco Diaz (Sales) - Account Executive
        GitHub: none
        FileSystem: Sales

Select persona (eng01/it01/sales01):

On selecting any persona, you see a basic menu,

[INFO] Using Docker-based GitHub MCP Server
[INFO] FileSystem folder ready: /home/dell/mcp_hyderabad1/
      persona_data/Engineering
[INFO] Database initialized: /home/dell/mcp_hyderabad1/
      persona_data/tickets.db
[INFO] MCP Client initialized for Alex Chen
[INFO] GitHub Access: read_write
[INFO] FileSystem Access: Engineering

Commands: <prompt> | persona | help | quit
Type 'help' for example prompts
=====
```

(a) Give command - persona, you see the personas listed along with GitHub and FileSystem access

```
=====
Available Personas:
=====
```

```
eng01: Alex Chen (Engineering) - Software Engineer
      GitHub: read_write
      FileSystem: Engineering
```

```
it01: Priya Nair (IT) - ITSM Analyst
      GitHub: read_only
      FileSystem: IT
```

```
sales01: Marco Diaz (Sales) - Account Executive
        GitHub: none
        FileSystem: Sales
```

Select persona (eng01/it01/sales01):

(b) Give command - help, you see a basic list of prompts

```
=====

EXAMPLE PROMPTS BY PERSONA:
=====

[ENGINEERING - eng01]
-----
    list repositories - List your GitHub repos
    read file - Read a file from GitHub
    create file - Create a new file in repository
    raise ticket - Raise an issue for IT

[IT - it01]
-----
    list repositories - List repos (read-only)
    list files - List files in IT folder
    read ticket - Read ticket details
    list tickets - Show all tickets
    resolve ticket - Close and resolve a ticket

[SALES - sales01]
-----
    list files - List files in Sales folder
    create proposal - Create proposal files
    raise ticket - Raise an issue for IT

=====
Commands: <prompt> | persona | help | quit
=====
```

(c) Give command - quit, the CLI closes

**Engineering Persona:** Select "eng01". Then give the following commands.

(a) list\_repositories

```
> list_repositories
```

```
[PROMPT] list_repositories
```

```
-----
```

```
[TASK] Unknown Task
```

```
[TOOLS] github
```

```
[MCP] Starting Docker container: docker run -i --rm ... ghcr.io  
/github/github-mcp-server
```

```
MCP Sending initialize request...
```

```
[MCP] Connected to GitHub MCP Server: github-mcp-server vv0  
.20.1
```

```
[MCP] Found 36 public repositories for vanivamshi
```

```
[MCP] Found 36 repositories in search result
```

```
=====
```

```
RESULTS:
```

```
=====
```

```
[OK] GITHUB SUCCESS:
```

```
{  
  "count": 36,  
  "repositories": [  
    {  
      "name": "Spotify-playlist-using-terraform",  
      "full_name": "vanivamshi/Spotify-playlist-using-terraform",  
      "private": false,  
      "description": "",  
      "url": "https://github.com/vanivamshi/Spotify-playlist-using-terraform"  
    },  
    {  
      "name": "LLM-agent-mcp",  
      "full_name": "vanivamshi/LLM-agent-mcp",  
      "private": false,  
      "description": "",  
      "url": "https://github.com/vanivamshi/LLM-agent-mcp"  
    },  
    {  
      "name": "Bigmart-sales-prediction",  
      "full_name": "vanivamshi/Bigmart-sales-prediction",  
      "private": false,  
      "description": "",  
      "url": "https://github.com/vanivamshi/Bigmart-sales-prediction"  
    },  
    {  
      "name": "Adversarial-Robustness-Analysis-of-BERT",
```

```

    "full_name": "vanivamshi/Adversarial-Robustness-Analysis-
      of-BERT",
    "private": false,
    "description": "",
    "url": "https://github.com/vanivamshi/Adversarial-
      Robustness-Analysis-of-BERT"
  },

```

... (truncated, total length: 8125 characters)

=====

### (b) read file

> read file

[PROMPT] read file

-----

[TASK] Unknown Task

[TOOLS] github, filesystem

[INPUT] Enter GitHub owner/username:

> vanivamshi

[INPUT] Enter repository name:

> LLM-agent-mcp

[INPUT] Enter file path (e.g., README.md):

> file.txt

[OK] GITHUB SUCCESS:

```

{
  "name": "",
  "path": "file.txt",
  "size": 11,
  "content": "how are you",
  "url": "repo://vanivamshi/LLM-agent-mcp/contents/file.txt",
  "sha": "81d97db467abde7c8ae570e1d8323ef1cf4405cf"
}

```

### (c) Create file

> create file

[PROMPT] create file

-----

[TASK] Unknown Task

[TOOLS] github, filesystem

[INPUT] Where to create file? (github/filesystem/both) [default  
: filesystem]:

> github

[INPUT] Enter GitHub owner/username:

> vanivamshi

[INPUT] Enter repository name:

> LLM-agent-mcp

[INPUT] Enter file path to create (e.g., new\_file.txt):

> world.c

[INPUT] Enter file content:

> hello world

[OK] GITHUB SUCCESS:

```
{
  "path": "world.c",
  "sha": "95d09f2b10159347eece71399a7e2e907ea3df4f",
  "url": "https://github.com/vanivamshi/LLM-agent-mcp/blob/main/world.c",
  "message": "File created successfully"
}
```

(d) fix bug

> fix bug

[PROMPT] fix bug

-----  
[TASK] Production Support

[TOOLS] github, filesystem

[INPUT] Enter GitHub owner/username:

> vanivamshi

[INPUT] Enter repository name:

> LLM-agent-mcp

[INPUT] Enter file path to fix (e.g., main.py):

> file.txt

[INFO] Reading existing file...

[INFO] Current file content (11 bytes):

-----  
how are you  
-----

[INPUT] Enter new/fixed content:

> weather is good

[INFO] Updated file content (15 bytes):

-----  
weather is good  
-----

[OK] GITHUB SUCCESS:

```
{
  "path": "file.txt",
  "sha": "",
  "url": "",
  "message": "File updated successfully",
  "updated_content": "weather is good"
}
```

## (e) raise ticket

```
> raise ticket
```

```
[PROMPT] raise ticket
```

```
-----
[TASK] Incident/Ticket Resolution
[WARNING] Task requires Priya Nair
[TOOLS] filesystem
[INPUT] Enter ticket/complaint title:
> password not working
[INPUT] Enter ticket/complaint description:
> My email password is not working. Not able to change it
[INPUT] Enter priority (Low/Medium/High) [default: Medium]:
> High
[INPUT] Enter category (optional):
>
[INFO] Ticket created: TKT-20251109100949-eng01
```

```
=====
RESULTS:
=====
```

```
[OK] FILESYSTEM SUCCESS:
{
  "id": 5,
  "ticket_id": "TKT-20251109100949-eng01",
  "title": "password not working",
  "description": "My email password is not working. Not able to
    change it",
  "status": "Open",
  "persona_id": "eng01",
  "persona_name": "Alex Chen",
  "team": "Engineering",
  "created_at": "2025-11-09 04:39:49",
  "updated_at": "2025-11-09 04:39:49",
  "resolved_at": null,
  "priority": "High",
  "category": null
}
```

**IT Persona:** Select "it01". Then give the following commands. Here "list\_repositories" and "read file" works similar to that demonstrated in Engineering persona.

## (a) file files - Lists files in local FileSystem

```
> list files
```

```
[PROMPT] list files
```

```
[TASK] Unknown Task
[TOOLS] filesystem
[FS] Listing files in: /home/dell/mcp/persona_data/IT
```

```
[OK] FILESYSTEM SUCCESS:
{
  "folder": "IT",
  "path": "/home/dell/mcp/persona_data/IT",
  "count": 2,
  "files": [
    {
      "name": "maintenance_script.sh",
      "type": "file",
      "size": 77,
      "path": "IT/maintenance_script.sh"
    },
    {
      "name": "incident_log.txt",
      "type": "file",
      "size": 102,
      "path": "IT/incident_log.txt"
    }
  ]
}
```

**(b) read ticket**

```
> read ticket
```

```
[PROMPT] read ticket
```

```
-----
[TASK] Incident/Ticket Resolution
[TOOLS] filesystem
```

```
=====
RESULTS:
=====
```

```
[OK] FILESYSTEM SUCCESS:
Ticket ID: TKT-20251109100949-eng01
Title: password not working
Status: Open
Priority: High
Updated at: 2025-11-09 04:39:49
```

```
Update History (1 updates):
```

```
-----
>>> LATEST UPDATE <<<
```

```
[1] Ticket created: password not working
    By: Alex Chen at 2025-11-09 04:39:49
```



## (c) list ticket

```
> list tickets
```

```
[PROMPT] list tickets
```

```
-----
```

```
[TASK] Incident/Ticket Resolution
```

```
[TOOLS] filesystem
```

```
=====
```

```
RESULTS:
```

```
=====
```

```
[OK] FILESYSTEM SUCCESS:
```

```
{
  "count": 5,
  "tickets": [
    {
      "id": 5,
      "ticket_id": "TKT-20251109100949-eng01",
      "title": "password not working",
      "description": "My email password is not working. Not
        able to change it",
      "status": "Open",
      "persona_id": "eng01",
      "persona_name": "Alex Chen",
      "team": "Engineering",
      "created_at": "2025-11-09 04:39:49",
      "updated_at": "2025-11-09 04:39:49",
      "resolved_at": null,
      "priority": "High",
      "category": null
    },
    {
      "id": 4,
      "ticket_id": "TKT-20251108152019-sales01",
      "title": "need helpers",
      "description": "people are required",
      "status": "Resolved",
      "persona_id": "sales01",
      "persona_name": "Marco Diaz",
      "team": "Sales",
      "created_at": "2025-11-08 09:50:19",
      "updated_at": "2025-11-08 09:59:09",
      "resolved_at": "2025-11-08 09:59:09",
      "priority": "Medium",
      "category": null
    }
  ],
  {
```

```

    "id": 3,
    "ticket_id": "TKT-

```

... (truncated, total length: 2296 characters)

(d) list open tickets

```
> list open tickets
```

```
[PROMPT] list open tickets
```

```
[OK] FILESYSTEM SUCCESS:
```

```

{
  "count": 2,
  "tickets": [
    {
      "id": 5,
      "ticket_id": "TKT-20251109100949-eng01",
      "title": "password not working",
      "description": "My email password is not working. Not
        able to change it",
      "status": "Open",
      "persona_id": "eng01",
      "persona_name": "Alex Chen",
      "team": "Engineering",
      "created_at": "2025-11-09 04:39:49",
      "updated_at": "2025-11-09 04:39:49",
      "resolved_at": null,
      "priority": "High",
      "category": null
    },
    {
      "id": 1,
      "ticket_id": "TKT-20251107223913-eng01",
      "title": "not able to access system",
      "description": "system access denied",
      "status": "Open",
      "persona_id": "eng01",
      "persona_name": "Alex Chen",
      "team": "Engineering",
      "created_at": "2025-11-07 17:09:13",
      "updated_at": "2025-11-07 17:09:13",
      "resolved_at": null,
      "priority": "High",
      "category": null
    }
  ],
  "filter": "Open"
}

```

(e) list resolved tickets

```
> list resolved tickets
```

```
[PROMPT] list resolved tickets
```

```
[OK] FILESYSTEM SUCCESS:
```

```
{
  "count": 3,
  "tickets": [
    {
      "id": 4,
      "ticket_id": "TKT-20251108152019-sales01",
      "title": "need helpers",
      "description": "people are required",
      "status": "Resolved",
      "persona_id": "sales01",
      "persona_name": "Marco Diaz",
      "team": "Sales",
      "created_at": "2025-11-08 09:50:19",
      "updated_at": "2025-11-08 09:59:09",
      "resolved_at": "2025-11-08 09:59:09",
      "priority": "Medium",
      "category": null
    },
    {
      "id": 3,
      "ticket_id": "TKT-20251108150731-eng01",
      "title": "no wifi",
      "description": "wifi not working",
      "status": "Resolved",
      "persona_id": "eng01",
      "persona_name": "Alex Chen",
      "team": "Engineering",
      "created_at": "2025-11-08 09:37:31",
      "updated_at": "2025-11-08 09:46:02",
      "resolved_at": "2025-11-08 09:46:02",
      "priority": "low",
      "category": "wifi"
    },
    {
      "id": 2,
      "ticket_id": "TKT-20251107224235-eng01",
      "
```

```
... (truncated, total length: 1385 characters)
```

(f) read open ticket

```
> read ticket TKT-20251109100949-eng01
```

```
[PROMPT] read ticket TKT-20251109100949-eng01
```

```
-----
```

[TASK] Incident/Ticket Resolution  
[TOOLS] filesystem

=====

RESULTS:

=====

[OK] FILESYSTEM SUCCESS:  
Ticket ID: TKT-20251109100949-eng01  
Title: password not working  
Status: Open  
Priority: High  
Updated at: 2025-11-09 04:39:49

Update History (1 updates):

-----

>>> LATEST UPDATE <<<  
[1] Ticket created: password not working  
By: Alex Chen at 2025-11-09 04:39:49

(g) modify ticket

> modify ticket

[PROMPT] modify ticket

-----

[TASK] Incident/Ticket Resolution  
[TOOLS] filesystem

[INFO] Current ticket: TKT-20251109100949-eng01  
Title: password not working  
Status: Open  
Priority: High

-----

[INPUT] Enter update text (or press Enter for status update only):

> password has been reset

[INPUT] Enter new priority (Low/Medium/High) or press Enter to keep current:

> High

[INFO] Ticket updated: TKT-20251109100949-eng01

=====

RESULTS:

=====

[OK] FILESYSTEM SUCCESS:  
Ticket ID: TKT-20251109100949-eng01  
Title: password not working  
Status: Open

Priority: High  
 Updated at: 2025-11-09 04:59:24

Update History (2 updates):

```
-----
[1] Ticket created: password not working
    By: Alex Chen at 2025-11-09 04:39:49
>>> LATEST UPDATE <<<
[2] password has been reset
    By: Priya Nair at 2025-11-09 04:59:24
-----
```

(h) close ticket

> close ticket

[PROMPT] close ticket

```
-----
[TASK] Incident/Ticket Resolution
[TOOLS] filesystem
[DEBUG] get_ticket: No ticket_id provided, getting most recent
        ticket for it01
[DEBUG] get_ticket: Found ticket: TKT-20251109100949-eng01
[DEBUG] Extracted ticket_id: TKT-20251109100949-eng01
[DEBUG] get_ticket: Querying for ticket_id: 'TKT
        -20251109100949-eng01'
[DEBUG] get_ticket: Found ticket: TKT-20251109100949-eng01
```

```
[INFO] Current ticket: TKT-20251109100949-eng01
Title: password not working
Status: Open
Priority: High
-----
```

```
[INPUT] Enter update text (or press Enter for status update
        only):
> issue resolved
[INPUT] Enter new priority (Low/Medium/High) or press Enter to
        keep current:
> High
[INFO] Ticket updated: TKT-20251109100949-eng01
```

```
=====
RESULTS:
=====
```

```
[OK] FILESYSTEM SUCCESS:
Ticket ID: TKT-20251109100949-eng01
Title: password not working
Status: Resolved
```

Priority: High  
 Updated at: 2025-11-09 05:05:33

Update History (3 updates):

```
-----
[1] Ticket created: password not working
    By: Alex Chen at 2025-11-09 04:39:49

[2] password has been reset
    By: Priya Nair at 2025-11-09 04:59:24
>>> LATEST UPDATE <<<
[3] issue resolved
    By: Priya Nair at 2025-11-09 05:05:33
```

**Sales Persona:** Select "sales01". Then give the following commands. Here "list\_repositories", "read file" and "raise ticket" works similar to that demonstrated in Engineering persona.

(a) create file

> create file

[PROMPT] create file

```
-----
[TOOLS] filesystem
[INPUT] Enter filename:
> abc.cpp
[INPUT] Enter content:
> cout<< "Hello" << endl;
[FS] Writing file: /home/dell/mcp_hyderabad1/persona_data/Sales
    /abc.cpp
```

```
=====
RESULTS:
=====
```

[OK] FILESYSTEM SUCCESS:

```
{
  "filename": "abc.cpp",
  "folder": "Sales",
  "path": "/home/dell/mcp_hyderabad1/persona_data/Sales/abc.cpp",
  "size": 23,
  "status": "created"
}
```